

Engineering Notes

Remote Uncorrelated Pilot Input Excitation Assessment for Unmanned Aircraft Aerodynamic Modeling

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I. Introduction

DURING flight test experiments of an unmanned aircraft, it is sometimes convenient and important for a human pilot to maintain active control. When operating the aircraft near its performance limits, for example, an attentive pilot must remain aware of the inputs and the resulting motion should the pilot need to recover from a stall or departure. This Engineering Note concerns a remote piloting technique that allows flight test engineers to obtain high-quality input/output data while the test pilot retains situational awareness and control of the aircraft motion. Specifically, the paper describes the development and evaluation of the remote uncorrelated pilot input (UPI) flight test technique (FTT) to excite aircraft motion for flight dynamic model identification. UPI is a modern technique that allows piloted excitation of the aircraft dynamics in all axes simultaneously [1–3]. The FTT is implemented by random excitation of the control surfaces with an effort to decorrelate the inputs while varying the input frequency and amplitude. To the best knowledge of the authors, this effort, originally published in Ref. [4], presents the first application of UPI with a ground-based pilot applying excitations remotely. The paper also describes a simulation-based training tool for improving pilot technique. The remote UPI FTT and training methodology presented in this paper apply widely to unmanned aerial vehicle (UAV) flight test aircraft and programs ranging in scale from small, low-cost, academic research to large commercial and military projects such as the Kratos XQ-58 Valkyrie. Considering a particular small, fixed-wing aircraft, for this paper we compare the remote UPI method with conventional methods involving manual and automated inputs. With the resulting input/output data, we first determine the model structure using equation-error methods (EEMs), then identify the model parameters using output-error methods (OEMs), and finally validate the model using independent data sets obtained with

different input waveforms. The modeling results obtained using data from remote UPI excitations compare favorably with the results obtained using more conventional methods.

The primary new contributions of this work are validation of the UPI technique for remotely piloted aircraft, documentation of training and best practices for remote UPI on a small UAV (sUAV), and comparison of a model developed with remote UPI to models developed with conventional piloted inputs and computer-generated automated inputs for a sUAV.

The remainder of the paper is organized as follows. Section II describes the aircraft system identification process. Section III describes the particular fixed-wing research aircraft considered here, the flight test experiment design, and remote UPI training methods. Section IV presents results from analysis of flight test data. Section V discusses conclusions based on the results.

II. Background

A. System Identification Process

Aircraft system identification seeks to develop an aerodynamic model, typically a set of differential or algebraic equations, that adequately describes the aircraft motion or response (outputs) to controls or perturbations (inputs). The aircraft system identification methodology used in this work, inspired by Ref. [5], is shown in Fig. 1. The process results in a high-quality nonlinear model that can accurately predict aircraft outputs for known inputs. System Identification Program for Aircraft (SIDPAC) was used for many components of this research, including experiment design, data processing, model structure determination, parameter estimation, and validation. SIDPAC was developed at NASA Langley Research Center, has been validated extensively, and is well documented in Ref. [5]. In general, the methods and discussion presented are intended for open-loop stable aircraft without feedback control. When feedback control is used, then automated excitations applied to control effector commands just before the actuator rate and position limits are required, to avoid high correlations in the data. More information and flight test examples related to this issue are provided in Refs. [5–9].

B. System Identification Excitation Inputs

For aircraft system identification, specific maneuvers are used to appropriately excite the aircraft natural dynamic motion and achieve high-quality modeling results from flight data. Details on input designs historically used for aircraft system identification are provided in Refs. [5,10–17]. Multistep inputs, including doublets, 1-2-1, and 3-2-1-1 inputs, are designed to be conducted near the aircraft short period and Dutch roll natural frequencies. The amplitude is chosen to obtain a high signal-to-noise ratio, without violating any of the modeling assumptions. Multistep inputs have the advantage of being simple to implement. A skilled pilot can implement these inputs directly and can vary the frequency of the input in real-time while observing the aircraft response. A computer can be programmed for highly repeatable automated maneuvers with *a priori* knowledge and the excitation frequency and amplitude may be adjusted in real-time from results [5]. Automated inputs are routinely referred to as programmable test inputs (PTIs) in practice. “Automated” and “PTI” terminology are used interchangeably in this paper.

Frequency sweeps and oscillatory inputs, such as the Schroeder-phased harmonic signal, can also excite a broad spectrum of frequencies. Oscillatory inputs are well suited for frequency domain analysis techniques and require minimal *a priori* knowledge of the system [5,17]. Frequency sweeps can be automated or implemented by a skilled pilot. While flight data from oscillatory inputs are not used for results in this paper, automated frequency sweeps were conducted

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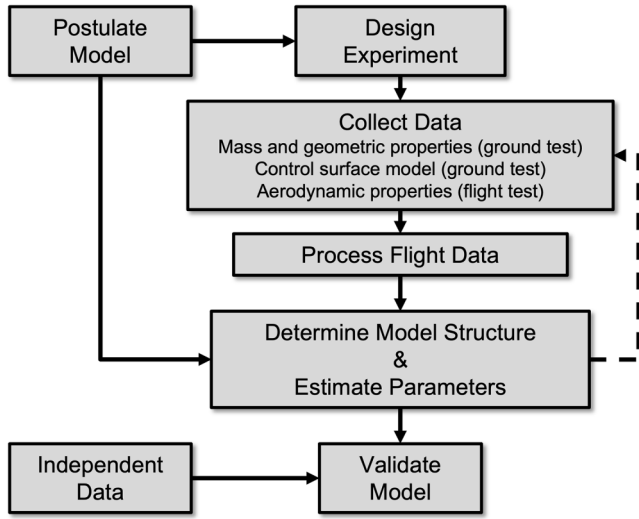


Fig. 1 Aircraft system identification process for a small, fixed-wing aircraft, adapted from Ref. [5].

during flight testing and the data were analyzed to evaluate the potential for aeroelastic phenomena.

Orthogonal phase-optimized multisine inputs [5–9] exercise multiple control surfaces exciting the natural aircraft longitudinal and lateral-directional motion concurrently. A multisine input is a summation of sinusoidal signals of varying frequency, amplitude, and phase angle. In the design of a multisine for aircraft system identification, the frequencies selected should cover a band that includes the frequencies associated with the dynamic modes of interest. The amplitudes can be selected to create an even power distribution for the frequencies of interest; however, they may also be tailored for specific modeling objectives. Multisine excitations offer the primary advantage of minimizing test time by exciting motion in all axes simultaneously while also designing all control surface excitations to be mutually orthogonal. Multisine input design does not require *a priori* information of the aircraft dynamics, but does require automated implementation.

A modern technique that maintains many of the advantages of multisines but does not require the engineering development effort to integrate automated excitation inputs is known as the UPI FTT. The foundation for the UPI technique was provided by Refs. [1–3] using simulation and manned aircraft flight data. Brandon and Morelli [3] state that UPI is a “piloted quasi-random input with varying frequency content and amplitude” that approximates “automated optimized multi-axis excitation inputs.” This efficient FTT reduces flight time required compared to single axis piloted inputs, while simultaneously allowing the pilot to control other aircraft states. UPI can be flown through stall and poststall regions because of the pilot’s ability to continuously correct with small offsets and subsequently recover the aircraft. A key disadvantage of UPI is the training required for the pilot and the fact that inputs will not be purely orthogonal as they are for a computer-generated multisine.

Use of remote UPI for unmanned aircraft system identification offers multiple benefits over classic piloted or automated system identification input techniques. Multistep inputs, including classical doublets, typically require a large number of repetitions to determine reasonable confidence intervals, and separate maneuvers are necessary for each control input, collectively leading to longer flight times. Multistep inputs may be piloted or automated, and proper implementation requires knowledge of the dynamic mode characteristics. Multisine inputs require automated inputs; therefore, some engineering development effort is necessary to implement appropriate computer-generated inputs for a new aircraft. Remote UPI is an appealing excitation method early in the UAV flight test program before automated inputs have been implemented, or when cost or programmatic speed is a concern because of the added effort to incorporate automated inputs. The UPI technique provides an effective and efficient piloted

multi-axis input when contrasted with doublets, multistep, or frequency sweep inputs [18]. In addition, remote UPI, similar to well-implemented multisine inputs, allows the pilot to remain in the control loop for state adjustment and upset recovery while characterizing stall effects, coupled dynamics, or interaction effects. Remote UPI is particularly beneficial for use in nonlinear flight regions where an autopilot is not designed or trusted to recover from departures [2]. The use of UPI as a system identification excitation input does require a pilot with some specialized training; however, there are instances when it may be the preferred FTT. Moreover, one of the contributions of this paper concerns preliminary pilot training.

C. Modeling Approach

A model is initially postulated from physical principles to aid in designing the experiment, and then the final model structure is determined based on experimental data. The aerodynamic model structure is founded on the fixed-wing aircraft translational and rotational kinematic equations and the force and moment dynamics equations derived in many classic flight mechanics textbooks [19–21].

The aerodynamic forces (X, Y, Z) and moments ($\mathcal{L}, \mathcal{M}, \mathcal{N}$) in the body axes are expressed in the nondimensional form

$$C_x = \frac{X}{\bar{q}S}, \quad C_y = \frac{Y}{\bar{q}S}, \quad C_z = \frac{Z}{\bar{q}S} \quad (1)$$

$$C_l = \frac{\mathcal{L}}{\bar{q}Sb}, \quad C_m = \frac{\mathcal{M}}{\bar{q}Sc}, \quad C_n = \frac{\mathcal{N}}{\bar{q}Sb} \quad (2)$$

to remove dependence on dynamic pressure and aircraft size. Aircraft system identification is accomplished by obtaining expressions for these nondimensional aerodynamic force and moment coefficients [5,22]. This research involved maneuvers with idle propulsion; thus thrust forces are omitted from the modeling and windmilling propeller effects are included in the aerodynamic model. Aerodynamic force coefficients are constructed from flight data by using mass (m), dynamic pressure ($\bar{q}$), and body-frame measurements of translational acceleration (a_x, a_y, a_z):

$$C_x = \frac{ma_x}{\bar{q}S} \quad (3)$$

$$C_y = \frac{ma_y}{\bar{q}S} \quad (4)$$

$$C_z = \frac{ma_z}{\bar{q}S} \quad (5)$$

The aerodynamic moment coefficients are estimated from flight data by combining the nondimensional moment equations shown in Eq. (2) and the rotational dynamics equations, leading to

$$C_l = \frac{1}{\bar{q}bS} \left(I_{xx}\dot{p} - I_{xz}(\dot{r} + pq) + qr(I_{zz} - I_{yy}) \right) \quad (6)$$

$$C_m = \frac{1}{\bar{q}cS} \left(I_{yy}\dot{q} + rp(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \right) \quad (7)$$

$$C_n = \frac{1}{\bar{q}bS} \left(I_{zz}\dot{r} - I_{xz}(\dot{p} - qr) + pq(I_{yy} - I_{xx}) \right) \quad (8)$$

Direct measurements of angular acceleration are unavailable, so these terms are obtained as smoothed derivatives of the angular rate measurements [5]. For this effort, aircraft flight dynamics equations of motion (EOM) are decoupled into longitudinal EOM and lateral-directional EOM by assuming that roll angle and angle of sideslip are small, as well as small perturbations from the reference flight condition [20].

III. Flight Test Experiments

A. Research Aircraft

The research aircraft tested was the My Twin Dream (MTD), a commercially available radio-controlled (RC), foam-structure, fixed-wing aircraft, shown in Fig. 2. The MTD mass and geometric properties are given in Table 1. The aircraft was electrically powered by a four-cell 14.8 V 6750 mAh lithium-polymer battery, with a counter-rotating twin propeller configuration. The MTD was instrumented with a Pixhawk flight computer for data collection, as well as an onboard cocomputer for automated input generation and for advanced control research. Each flight test was controlled using a Spektrum 20-channel RC transmitter. Documentation of the instrumentation, cocomputer setup, and data processing techniques is available in Ref. [23].



Fig. 2 The MTD fixed-wing research aircraft.

Table 1 MTD mass and geometric properties

Property	Symbol	Value	Units
Mass	m	0.211	slug
Mean aerodynamic chord	$\bar{c}$	0.833	ft
Projected wing span	b	5.91	ft
Projected wing area	S	4.92	ft ²
Roll moment of inertia	I_{xx}	0.2163	slug · ft ²
Pitch moment of inertia	I_{yy}	0.1823	slug · ft ²
Yaw moment of inertia	I_{zz}	0.3396	slug · ft ²
Product of inertia	I_{xz}	0.0364	slug · ft ²

B. Experiment Design

The experiment for this effort was designed to develop a nonlinear model exclusively from flight test data without an *a priori* model from analytical techniques, computational fluid dynamics, or data collected with wind tunnel experimentation. A straightforward experiment setup was desired with pilot flown inputs compared to complementary automated inputs produced using an onboard cocomputer. Additionally, this effort aimed to compare classic techniques to the remote UPI technique flown from a third person view-point (view of the remote pilot from the ground) for a remotely piloted aircraft. For simplicity and clarity of comparison, basic FTTs were selected with a sufficient number of test points to produce reasonable confidence intervals. All maneuvers were initiated in a trimmed glide and remained near the relatively steady glide condition. Models developed from both piloted doublet and remote UPI maneuvers were compared with both PTI doublet and multisine maneuvers.

Flight test objectives included developing a nonlinear aerodynamic model for the normal operating range that, when linearized, captured the short period dynamics, Dutch roll dynamics, and roll mode time constant. FTTs such as elevator doublets, rudder doublets, and bank-to-bank aileron rolls (-45° to 45°) were chosen based on the ability of the pilot to make inputs without onboard sensory feedback, including sight-picture (real or simulated nose camera view) and acceleration cues normally experienced in the cockpit. Rudder doublets followed by aileron 1-2-1 inputs were combined into a single maneuver (referred to as a lateral-directional “doublet” for brevity) to capture effects from both control surfaces in a continuous data set. This permits parameter estimation for lateral-directional motion using the OEM with a single flight maneuver. Example piloted doublet and 1-2-1 inputs are shown in Fig. 3.

The pulse width of elevator and rudder doublets was initially determined in flight by the pilot conducting an impulse maneuver flown overhead, estimating the period of response visually, then inputting that period on subsequent doublet maneuvers. This proved to be an efficient and effective technique for designing sUAV doublet inputs. Automated elevator and rudder doublet input frequencies were selected to match the estimates of short period and Dutch roll frequencies from piloted maneuvers. Automated bank-to-bank roll aileron inputs were performed using the onboard estimate of bank angle provided by the Pixhawk EKF to autonomously reverse aileron input direction at $\pm 45^\circ$ of bank. Each doublet maneuver was intended to have at least 2 s of trimmed, steady flight before initiating the control input. Following the maneuver inputs, several seconds of free aircraft response were recorded while the aircraft returned to its original trimmed condition.

As described in Sec. II.B, orthogonal phase optimized multisine inputs were designed to develop a comprehensive aircraft dynamic model around a nominal flight condition that described nonlinear aerodynamic phenomena and control surface interaction effects. Three-axis multisine inputs were designed for a range of 0.1–2.0 Hz using SIDPAC [5], which covers the frequency range of the dynamic modes of interest for the MTD aircraft. The multisine inputs

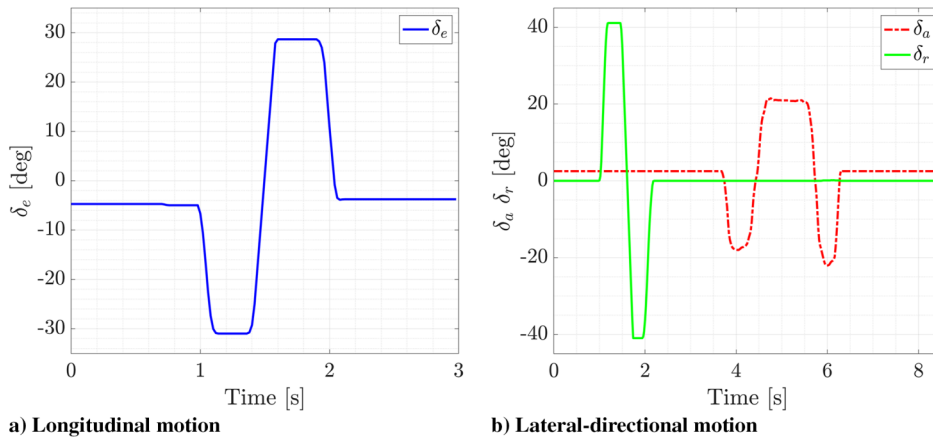


Fig. 3 Select piloted doublet inputs for system identification.

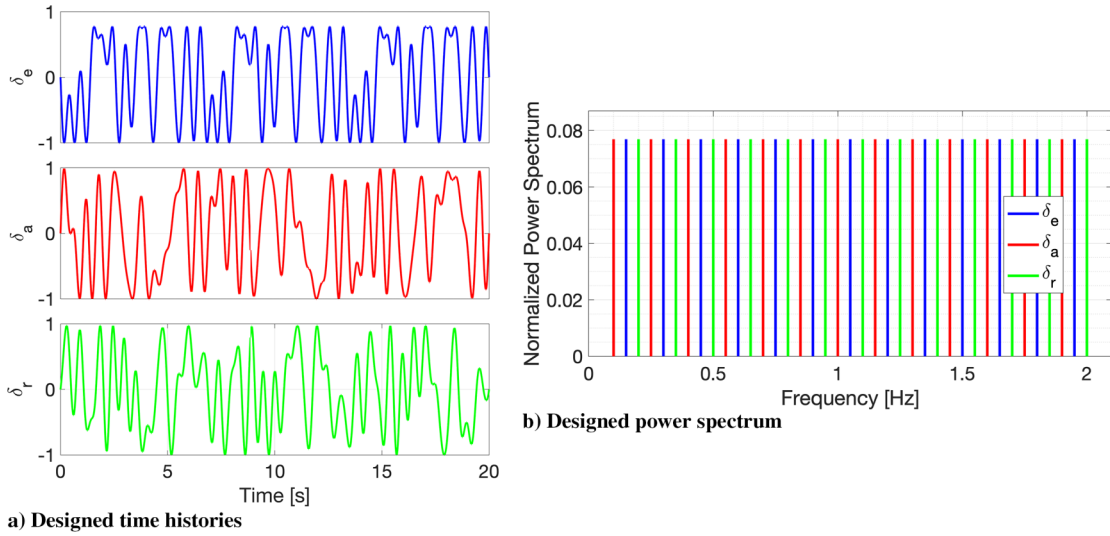


Fig. 4 Designed multisine control inputs.

were designed for the three aerodynamic control inputs (elevator, aileron, and rudder) with a uniform power spectrum and 39 total discrete frequencies and phase angles. The designed time histories and power spectrum are shown in Fig. 4 and the designed harmonics are listed in Table 2. As with the square-wave inputs, the amplitude of the multisine PTI inputs was normalized to unity and then adjusted in real-time by the pilot with a knob from 0 to 100% of available control surface deflection angle.

Implementation of multisine inputs and the remote UPI technique is discussed further in the following sections. Each multisine and remote UPI maneuver was intended to have at least 2 s of trimmed, steady flight followed by 30 s of excitation and flight data collection.

Automated 3-2-1-1 aircraft response data were used for model validation, in addition to a three-axis multisine with an independently designed and different waveform. Use of both square wave and rounded input maneuvers for validation allowed a fair comparison

of model predictions from models developed from each maneuver type. The automated elevator, rudder, and aileron 3-2-1-1 input frequencies were selected to correspond with the initial estimates of short period and Dutch roll frequencies from piloted maneuvers based on guidance in Refs. [5,22].

The number of planned maneuvers was sufficient to allow the removal of maneuvers with extraneous pilot inputs in an unplanned axis, or where atmospheric turbulence was observed in the trim data or response data. Atmospheric turbulence was apparent when aircraft motion appeared in the data with no corresponding control inputs. In a typical sUAV flight test profile, approximately one-third of the maneuvers contain undesirable data traits that are visible in the initial data preparation. Additionally, according to Ref. [5], at least five of each individual maneuver at each different flight condition should be included so that corrupted maneuvers can be removed. The planned test matrix is shown in Table 3.

C. Remote UPI Training

Two pilots performed remote UPI for data collection with the test aircraft for this effort. The primary test pilot was an experienced RC pilot and a trained flight test professional, but had no prior training or experience with the UPI technique. The second pilot was a novice RC pilot with no test training. The pilots reviewed Refs. [1–3] for background information on UPI. The experienced pilot accomplished initial remote UPI training during three practice flights performing 15 remote UPI maneuvers, and then compared remote UPI time histories that were collected to multisine time histories. Guided by the familiarity gained during the preliminary flight tests, a simulator-based training tool was developed to improve the pilot's ability to generate remote UPI excitations. The simulator concentrated on the input signal character and did not include an aircraft flight model. Training objectives included focusing the frequency content of inputs within the frequency band of common aircraft dynamic modes and decreasing correlation between control excitation signals.

Table 2 Model development orthogonal phase-optimized multisine harmonics ($T = 20$ s)

f_{δ_e} , Hz	ϕ_{δ_e} , rad	f_{δ_a} , Hz	ϕ_{δ_a} , rad	f_{δ_r} , Hz	ϕ_{δ_r} , rad
0.15	+1.35	0.10	-0.47	0.20	-3.09
0.30	+1.20	0.25	+1.29	0.35	-1.55
0.45	-0.59	0.40	-2.30	0.50	+3.10
0.60	-2.02	0.55	-2.25	0.65	-1.72
0.75	+0.27	0.70	-0.49	0.80	+0.43
0.90	+1.96	0.85	+3.02	0.95	+1.38
1.05	-1.93	1.00	+0.83	1.10	-2.25
1.20	-0.75	1.15	+2.95	1.25	-2.77
1.35	-2.06	1.30	+2.31	1.40	+1.94
1.50	+0.33	1.45	+1.63	1.55	+1.53
1.65	+0.14	1.60	-1.74	1.70	-2.61
1.80	+1.21	1.75	-2.81	1.85	+1.47
1.95	-0.57	1.90	+3.13	2.00	-0.75

Table 3 Planned test matrix

Maneuver	Control input	Input type	Runs	Duration, s
Longitudinal pilot doublet	Elevator doublet	Pilot	30	3.5
Longitudinal PTI doublet	Elevator doublet	PTI	30	4.5
Longitudinal PTI multistep	Elevator 3-2-1-1	PTI	5	11
Lateral-directional pilot doublet	Rudder doublet to aileron 1-2-1	Pilot	30	10
Lateral-directional PTI doublet	Rudder doublet to aileron 1-2-1	PTI	30	10
Lateral-directional PTI multistep	Rudder 3-2-1-1 to aileron 3-2-1-1	PTI	5	11
Remote UPI	3-axis UPI	Pilot	15	30
PTI multisine	3-axis multisine	PTI	15	30
PTI independent multisine	3-axis multisine	PTI	5	30

The simulation tool was designed using MATLAB® and Simulink [24]. As with analysis of flight data presented in the next section, the spectrum of the control signals was calculated using a high-accuracy Fourier transform technique described in Refs. [5,25] and available in SIDPAC, which allows using an arbitrary frequency resolution. The RC transmitter used for flight tests was connected to the simulation environment with a Spektrum WS2000 Wireless Simulator USB Dongle.

Pilots practiced remote UPI maneuvers in simulation and near-real-time input signal analysis was displayed, as shown in Fig. 5, in a matrix plot of control signal time traces (left), frequency content (center), and pairwise cross plots with a numerical correlation coefficient (r) values below 0.3, and evenly distributing the control signal data throughout the cross plots for the full range of inputs. The training technique that produced rapid improvement was to perform a three-axis maneuver and subsequently focus on improving one axis at a time. Performing UPI on each axis individually was considered, but determined to provide negative training because pilot muscle memory reduced inputs on the off-axes in subsequent maneuvers. Both pilots trained on the simulator until achieving consistent inputs with the above objectives in fewer than 10 repetitions. As compared to manned aircraft inceptors or a simulator joystick inceptor, the experienced pilot commented that the returning force of the RC transmitter stick springs was significant and keeps the control sticks

centered when the pilot applied force is relatively low. This required additional concentration to generate inputs for the full range of motion. Remote UPI data were collected in flight with both pilots following this training.

D. Remote UPI in Flight

Several challenges were observed regarding the use of the remote UPI technique in flight. First, some experience was required to generate random excitations while maintaining the aircraft upright in its normal operating envelope. The novice pilot commented that it was far more difficult to perform remote UPI in flight than in the training simulator because of the need to constantly correct the attitude. Second, the lack of a horizon reference line from the third-person (ground) view made trimming the aircraft and maintaining a roughly trim condition throughout the maneuver difficult. However, this did not appear to adversely impact data quality. This challenge was overcome by occasionally referencing the attitude display on the ground station telemetry data, and could be further aided by a future nose camera or first-person-view video feed. Third, the approach to stall or stall break was not readily apparent due to a lack of physiological cues, including acceleration or auditory (airspeed) changes. The lack of some form of feedback made it more challenging to intentionally approach stall or prevent it when desired. This poor awareness could be improved with visual or auditory feedback of telemetered airspeed to the pilot. Fourth, the pilots' initial tendency was to generate rudder and aileron inputs that were well decorrelated

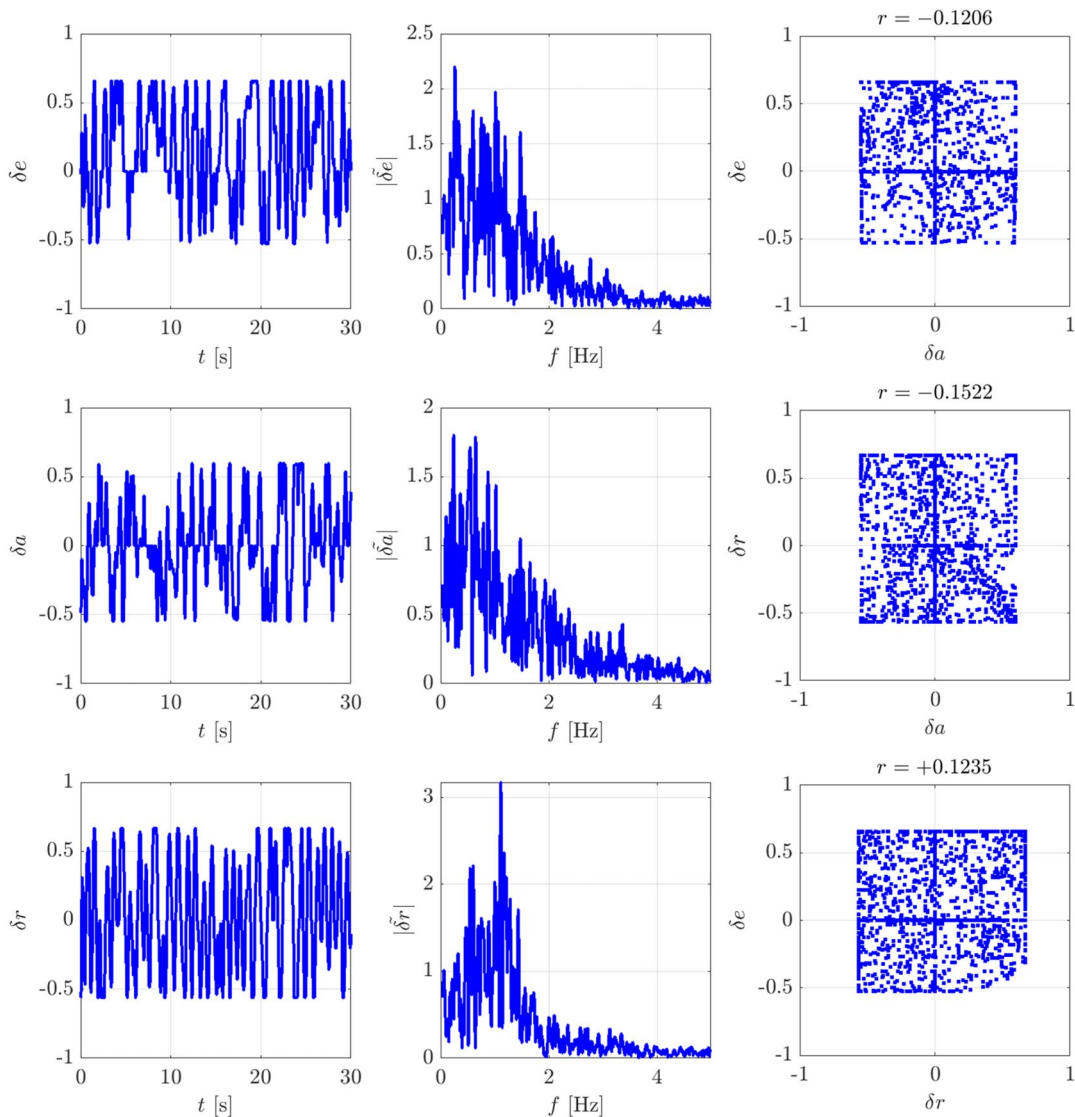


Fig. 5 Remote UPI simulator training output.

in flight, but elevator inputs were of insufficient amplitude to use for modeling. The experienced pilot commented initially that it was straightforward to randomly generate the left–right motion on both RC sticks, but adding the forward–aft input on the right stick for pitch inputs required additional concentration while maintaining the nominal attitude.

With training and experience, the pilots produced uncorrelated inputs with broad frequency content while remaining in the desired operating envelope. The resulting time trace and frequency content of a remote UPI maneuver from flight data are compared with that of an automated multisine in Figs. 6 and 7, respectively. The flight test data

of the PTI multisine differ from the designed multisine with uniform power distribution because the pilot made minor adjustments during the maneuver to remain within the desired aircraft attitude limits, and because of the random nature of the amplitudes and frequencies for a piloted input.

A plot matrix displaying correlation between control excitations, following the example of [26], is shown in Fig. 8 for a multisine and a remote UPI maneuver. The explanatory variables are labeled along the diagonal, pairwise correlations are shown in the lower triangle, and corresponding pairwise cross plots are shown diagonally opposite in the upper triangle.

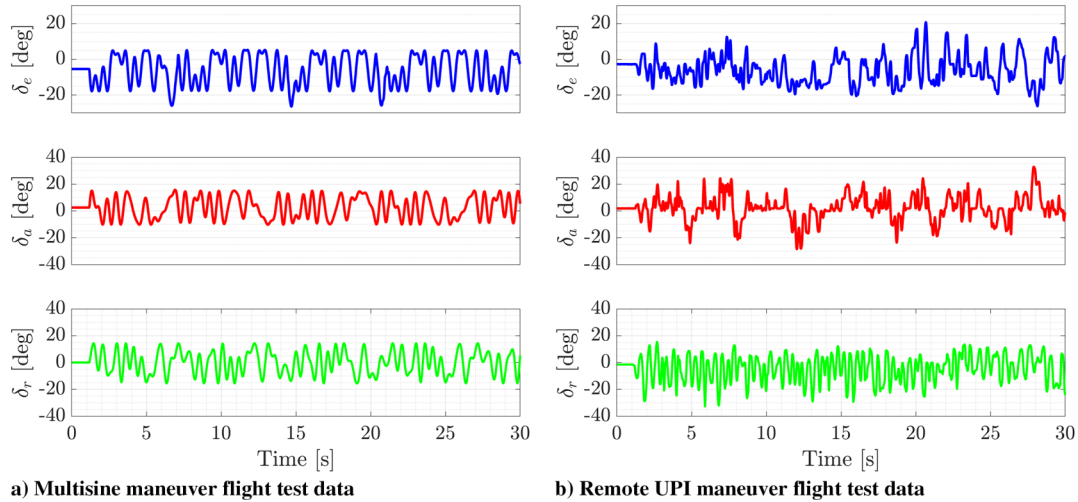


Fig. 6 Comparison of control input time histories for multisine and remote UPI maneuvers.

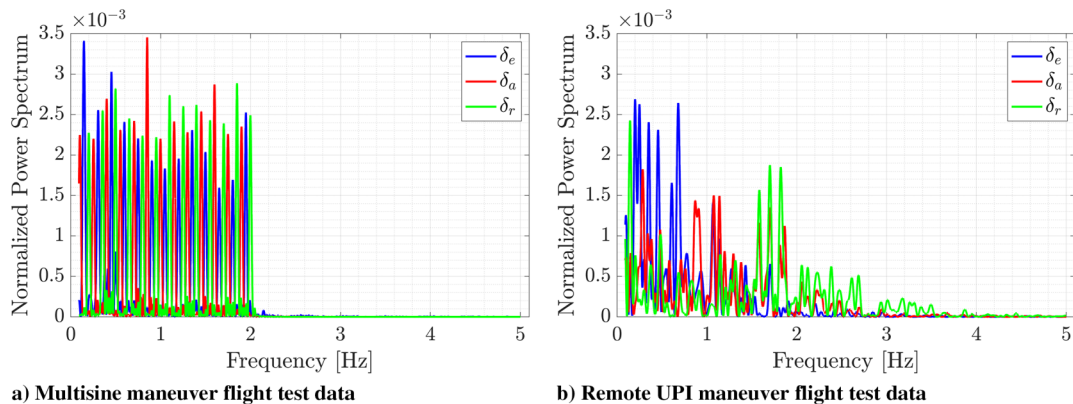


Fig. 7 Comparison of power spectrum of multisine and remote UPI maneuvers.

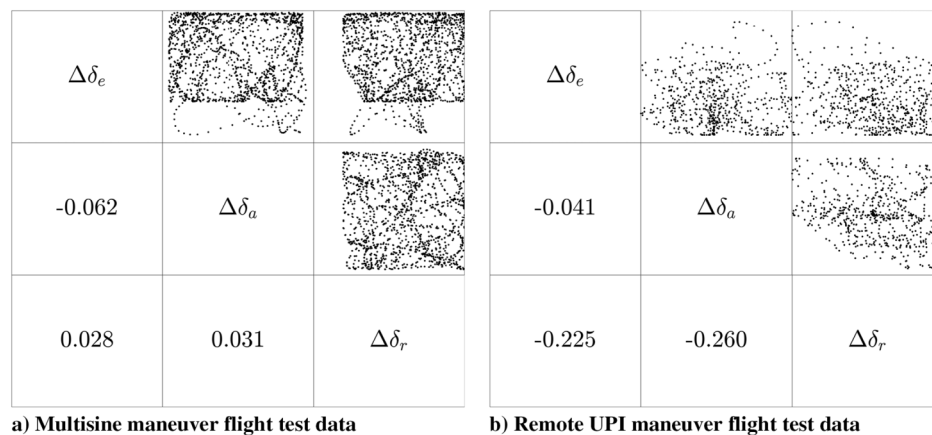


Fig. 8 Comparison of control input pairwise cross-correlation coefficients and cross plots for multisine and remote UPI maneuvers.

The flight test data comparing multisine and remote UPI excitations indicate that the simulator training was successful and the remote UPI flight data were well-suited for system identification analysis. The modeling utility of remote UPI maneuvers compared to other piloted and automated input maneuvers is explored in the next section.

IV. Results

Comparison of the modeling results from the four excitation techniques (remote pilot doublets, PTI doublets, remote UPI, and PTI multisines) is presented here. Data analysis was conducted in MATLAB® [24] with use of analytical tools developed by the authors and those provided in SIDPAC [5].

A. Model Structure Determination

For this research, both stepwise regression (SWR) [5,27] and multivariate orthogonal function (MOF) modeling [5,8,28] were employed for model structure identification. SWR was run with manual user input following the algorithm described in Refs. [5,27]. The explanatory variables used to describe the aerodynamic force and moment coefficients for this research were relative air flow states V_t , α , β , the nondimensional angular rate components in body axes

$$\hat{p} = \frac{pb}{2V_t}, \quad \hat{q} = \frac{q\bar{c}}{2V_t}, \quad \hat{r} = \frac{rb}{2V_t} \quad (9)$$

and the control surface angular deflections δ_a , δ_e , and δ_r . Candidate regressors included linear and nonlinear combinations of explanatory variables in all combinations, up to the third order. Model structure determination was initially performed with the combined pilot and PTI doublet maneuvers with select longitudinal and lateral-directional results from MOF modeling shown in Fig. 9, similar to the analysis presented in Ref. [29]. The candidate regressors are shown based on the number of maneuvers for which these regressors made a significant contribution to model performance. If the candidate regressor was selected in at least half of the maneuver data sets, as shown by the vertical line, it was included in the model structure for parameter estimation analysis. The model structure determination process was repeated with the remote UPI and multisine maneuvers, which reaffirmed the selected modeling terms. Analyst insight was also used to combine results from MOF and modified SWR to prioritize linear terms and lower-order terms. The same model structure was used to fit models for all input types to enable comparison of the individual parameters in each respective model. Following parameter estimation with equation-error and OEMs, terms such as $C_{l(p^3)}$ shown above the threshold in Fig. 9 were removed, which had an aggregate median and mean near zero with nearly equal runs having positive and negative parameter estimates. Additional terms removed included $C_{z(\alpha^2)}$, $C_{z(\delta_e)}$, $C_{m(\alpha^2)}$, $C_{y(\alpha^2)}$, $C_{y(\beta^2)}$, $C_{y(\beta^3)}$, $C_{l(p^3)}$, C_{n_p} ,

Table 4 Summary of validity range and nominal values for flight data

State	Units	5th percentile	Mean	95th percentile
α	deg	-3.1	4.0	11.9
β	deg	-15.0	-1.4	12.5
p	deg/s	-82.9	0.5	83.6
q	deg/s	-45.9	6.3	64.7
r	deg/s	-61.0	-1.1	61.6
δ_e	deg	-19.6	-5.9	3.9
δ_a	deg	-10.0	2.4	15.1
δ_r	deg	-14.2	0.2	14.2
V_t	ft/s	36.0	51.5	69.9

Total of 95 maneuvers with 938.9 s of data (longitudinal and lateral-directional combined).

$C_{n(\beta^2)}$, and $C_{n(\beta^3)}$. All regressors obtained for this aircraft were also part of the generic nonlinear aerodynamic model structure presented in Ref. [29].

The final identified model structure was

$$C_x = C_{x_\alpha} \alpha + C_{x_{\delta_e}} \delta_e + C_{x_{\alpha^2}} \alpha^2 + C_{x_o} \quad (10)$$

$$C_z = C_{z_\alpha} \alpha + C_{z_o} \quad (11)$$

$$C_m = C_{m_\alpha} \alpha + C_{m_q} \hat{q} + C_{m_{\delta_e}} \delta_e + C_{m_{\alpha^3}} \alpha^3 + C_{m_o} \quad (12)$$

$$C_y = C_{y_\beta} \beta + C_{y_p} \hat{p} + C_{y_r} \hat{r} + C_{y_{\delta_a}} \delta_a + C_{y_{\delta_r}} \delta_r + C_{y_o} \quad (13)$$

$$C_l = C_{l_\beta} \beta + C_{l_p} \hat{p} + C_{l_r} \hat{r} + C_{l_{\delta_a}} \delta_a + C_{l_o} \quad (14)$$

$$C_n = C_{n_\beta} \beta + C_{n_p} \hat{p} + C_{n_r} \hat{r} + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r + C_{n_o} \quad (15)$$

The model structure and parameter estimates presented in the next section provide regional models valid in the normal operating range, to include stall onset. They do not capture α - β coupling, poststall effects, or spin dynamics. All individual maneuvers used for data analysis for this paper encompassed similar velocities, angles of attack, angles of sideslip, and body rates. A summary of the data set is shown in Table 4, which includes the 5th percentile, mean, and 95th percentile for the states and control deflections used for modeling, as well as the velocity.

B. Model Parameter Estimation

For this paper, parameter estimation results using the time-domain EEM were used as initial inputs for the nonlinear solver in

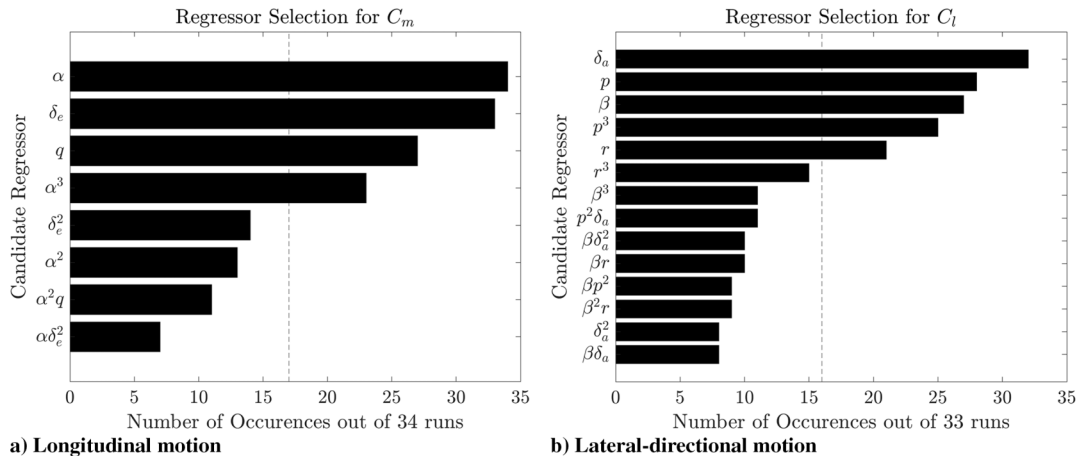


Fig. 9 Select results from multivariate orthogonal function modeling using doublet maneuvers.

the time-domain OEM. The EEM analysis in this paper uses a least-squares regression to minimize the difference between the nondimensional aerodynamic force and moment components computed from flight test data and those predicted by the mathematical model. The method is algebraic, requiring no numerical integration of the postulated dynamics, and is therefore computationally efficient. The OEM is based on a numerical integration approach to determine unknown parameters, and it is thus a more computationally intensive nonlinear optimization problem. The OEM minimizes the weighted sum of squared differences between measured aircraft outputs and modeled outputs computed through numerical integration of the proposed dynamic model [5]. For a correct model structure, the maximum likelihood parameter estimates will approach the true values as the amount of data increases. An estimate of the minimum variance in individual parameter estimates is given by the computed Cramér–Rao lower bound, and the estimated parameter covariance approaches this lower bound as the amount of data increases [30]. The OEM accounts for zero-mean, white, Gaussian measurement noise, but it does not allow for process noise, i.e., for disturbances such as atmospheric turbulence [31]. The OEM is used for final parameter estimates for this effort using input/output data that were collected in calm wind conditions (mean wind speed at or below 1 knot).

Initially, the OEM for the long-duration maneuvers (multisines and remote UPI) diverged for longitudinal modeling when initial parameter estimates were selected from their respective EEM results. It was determined that the test aircraft has an unstable phugoid mode. Unstable system dynamics complicate analysis using the OEM as the numerical integration can diverge, precluding maximum likelihood cost function convergence. An algorithm was developed to iteratively truncate the maneuver length until the OEM converged. Once a converged solution to the OEM was obtained for the truncated maneuver, those parameter estimates were used to re-initialize the

OEM while iteratively increasing the length of data used from the maneuver. This technique is described in Ref. [22] for OEM with unstable systems and was effective for this effort. The OEM was used to analyze the data from each run independently. Select parameter estimate results are shown in Fig. 10 with individual run uncertainty bars showing the Cramér–Rao bounds accounting for colored residuals [5,30]. These results show that parameter estimates and confidence intervals are comparable for automated multisine input maneuvers and remote UPI maneuvers. The mean value of the set of parameter estimates obtained from analyzing different maneuvers was taken as the final value. The uncertainty of the final parameter estimates, σ , was determined from the standard deviation for the data set. This uncertainty value σ was used to determine a 95% confidence interval approximated by $\pm 2\sigma$, as shown by the green-shaded region in Fig. 10.

Final parameter estimates for the longitudinal and lateral-directional models developed from the various inputs are shown in Tables 5 and 6, respectively. A comparison of parameter estimates from the models derived from various inputs, with their respective uncertainty quantified by the data set standard deviation, is shown in Figs. 11 and 12. The parameter estimates from each of the four input types are in relative agreement for individual parameter estimates. As discussed in Sec. III.D, initial remote UPI maneuvers did not have sufficient longitudinal content to be used for analysis, resulting in fewer maneuvers used to generate the longitudinal remote UPI model than the lateral-directional remote UPI model. This also resulted in larger aggregate parameter uncertainty for the longitudinal remote UPI model parameter estimates.

C. Model Validation

Model validation was accomplished by applying the models identified using flight data from maneuvers with various input types to

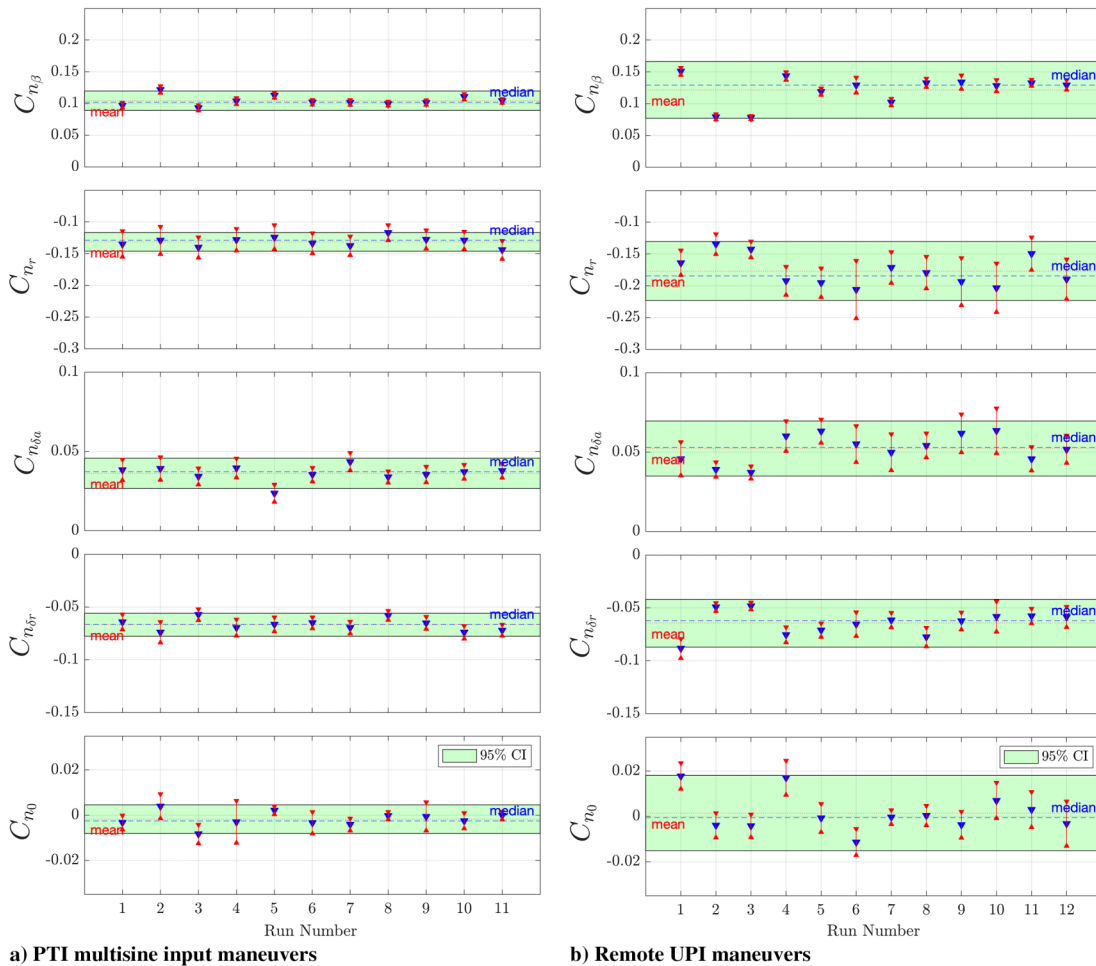


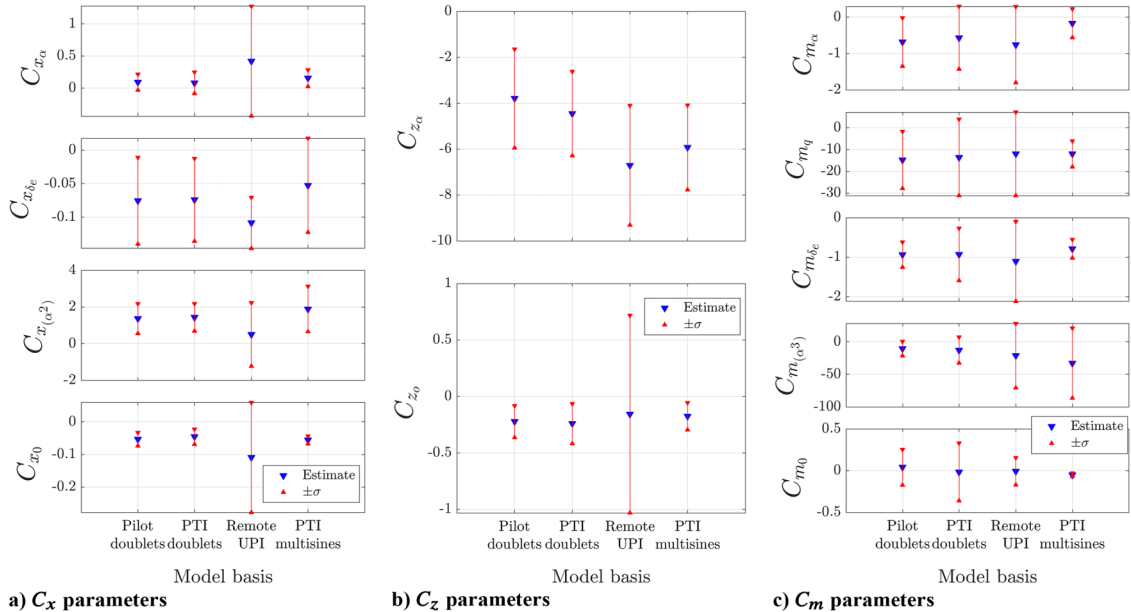
Fig. 10 Select lateral-directional parameter estimate comparison using the time-domain output-error method.

Table 5 Longitudinal parameter estimates

Parameter	Piloted doublets, $\hat{\theta} \pm \sigma$	PTI doublets, $\hat{\theta} \pm \sigma$	Remote UPI, $\hat{\theta} \pm \sigma$	PTI multisines, $\hat{\theta} \pm \sigma$
C_{x_a}	+0.095 $\pm$ 0.062	+0.082 $\pm$ 0.082	+0.420 $\pm$ 0.427	+0.156 $\pm$ 0.064
$C_{x_{\delta_e}}$	-0.076 $\pm$ 0.032	-0.074 $\pm$ 0.031	-0.108 $\pm$ 0.019	-0.052 $\pm$ 0.035
$C_{x_{(\alpha^2)}}$	+1.376 $\pm$ 0.405	+1.434 $\pm$ 0.375	+0.506 $\pm$ 0.863	+1.891 $\pm$ 0.618
C_{x_0}	-0.053 $\pm$ 0.010	-0.045 $\pm$ 0.012	-0.109 $\pm$ 0.084	-0.056 $\pm$ 0.006
C_{z_a}	-3.794 $\pm$ 1.076	-4.456 $\pm$ 0.919	-6.706 $\pm$ 1.301	-5.928 $\pm$ 0.919
C_{z_0}	-0.223 $\pm$ 0.070	-0.241 $\pm$ 0.088	-0.157 $\pm$ 0.438	-0.176 $\pm$ 0.060
C_{m_a}	-0.008 $\pm$ 0.021	-0.006 $\pm$ 0.017	-0.002 $\pm$ 0.015	-0.001 $\pm$ 0.006
C_{m_q}	-0.684 $\pm$ 0.333	-0.568 $\pm$ 0.430	-0.756 $\pm$ 0.523	-0.169 $\pm$ 0.198
$C_{m_{\delta_e}}$	-14.770 $\pm$ 6.548	-13.611 $\pm$ 8.763	-12.001 $\pm$ 9.586	-11.988 $\pm$ 2.988
$C_{m_{(\alpha^3)}}$	-0.936 $\pm$ 0.160	-0.928 $\pm$ 0.331	-1.107 $\pm$ 0.502	-0.788 $\pm$ 0.116
C_{m_0}	-10.635 $\pm$ 5.428	-12.801 $\pm$ 9.860	-21.235 $\pm$ 24.671	-32.831 $\pm$ 26.829

Table 6 Lateral-directional parameter estimates

Parameter	Piloted doublets, $\hat{\theta} \pm \sigma$	PTI doublets, $\hat{\theta} \pm \sigma$	Remote UPI, $\hat{\theta} \pm \sigma$	PTI multisines, $\hat{\theta} \pm \sigma$
C_{y_β}	-0.404 $\pm$ 0.029	-0.410 $\pm$ 0.060	-0.417 $\pm$ 0.090	-0.388 $\pm$ 0.029
C_{y_p}	+0.960 $\pm$ 0.192	+0.700 $\pm$ 0.136	+0.656 $\pm$ 0.127	+0.440 $\pm$ 0.087
C_{y_r}	+0.320 $\pm$ 0.086	+0.341 $\pm$ 0.083	+0.108 $\pm$ 0.082	+0.088 $\pm$ 0.063
$C_{y_{\delta_a}}$	+0.311 $\pm$ 0.047	+0.249 $\pm$ 0.048	+0.272 $\pm$ 0.053	+0.187 $\pm$ 0.027
$C_{y_{\delta_r}}$	+0.123 $\pm$ 0.013	+0.160 $\pm$ 0.034	+0.154 $\pm$ 0.026	+0.148 $\pm$ 0.013
C_{y_0}	-0.027 $\pm$ 0.013	-0.008 $\pm$ 0.026	-0.025 $\pm$ 0.035	-0.006 $\pm$ 0.014
C_{l_β}	-0.051 $\pm$ 0.006	-0.041 $\pm$ 0.010	-0.062 $\pm$ 0.018	-0.041 $\pm$ 0.009
C_{l_p}	-0.719 $\pm$ 0.098	-0.565 $\pm$ 0.089	-0.599 $\pm$ 0.070	-0.482 $\pm$ 0.055
C_{l_r}	+0.159 $\pm$ 0.034	+0.112 $\pm$ 0.038	+0.184 $\pm$ 0.081	+0.091 $\pm$ 0.049
$C_{l_{\delta_a}}$	-0.247 $\pm$ 0.027	-0.217 $\pm$ 0.036	-0.247 $\pm$ 0.040	-0.203 $\pm$ 0.023
C_{l_0}	+0.011 $\pm$ 0.005	+0.012 $\pm$ 0.003	+0.008 $\pm$ 0.004	+0.011 $\pm$ 0.003
C_{n_β}	+0.117 $\pm$ 0.007	+0.120 $\pm$ 0.018	+0.121 $\pm$ 0.022	+0.104 $\pm$ 0.008
C_{n_r}	-0.197 $\pm$ 0.021	-0.196 $\pm$ 0.035	-0.177 $\pm$ 0.023	-0.131 $\pm$ 0.007
$C_{n_{\delta_a}}$	+0.034 $\pm$ 0.006	+0.037 $\pm$ 0.005	+0.052 $\pm$ 0.009	+0.036 $\pm$ 0.005
$C_{n_{\delta_r}}$	-0.054 $\pm$ 0.005	-0.072 $\pm$ 0.013	-0.065 $\pm$ 0.011	-0.067 $\pm$ 0.005
C_{n_0}	+0.003 $\pm$ 0.004	-0.001 $\pm$ 0.005	+0.001 $\pm$ 0.008	-0.002 $\pm$ 0.003

**Fig. 11** Comparison of longitudinal parameter estimates for various input-derived models.

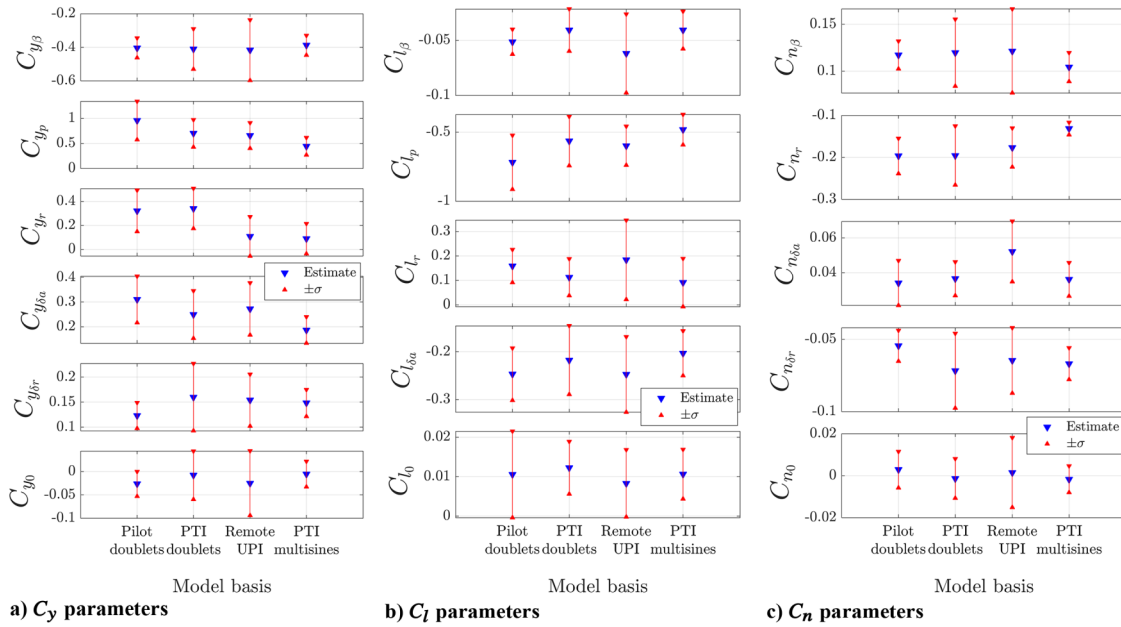


Fig. 12 Comparison of lateral-directional parameter estimates for various input-derived models.

predict flight data that were not used in the modeling process. For quantitative analysis, similar to Ref. [32], goodness of fit (GOF) and the Theil inequality coefficient (TIC) are determined for each model relative to two validation maneuvers. GOF is calculated from

$$\text{GOF} = 1 - \frac{(z - y)^T(z - y)}{(z - z_0)^T(z - z_0)} \quad (16)$$

where z is the measured flight test data for each output, z_0 is the initial value of the output in the data set, and y is the model predicted output. Similarly, TIC is calculated from

$$\text{TIC} = \frac{\sqrt{\frac{1}{N}(z - y)^T(z - y)}}{\sqrt{\frac{1}{N}z^Tz} + \sqrt{\frac{1}{N}y^Ty}} \quad (17)$$

A GOF value of 1 indicates a perfect model fit, a TIC value of 0 indicates a perfect model fit, and a TIC below 0.3 is considered good agreement between validation data and model predicted outputs [22].

The nonlinear model developed for each of the four input types is validated against separate input maneuvers, a 3-2-1-1 automated input, and an independently designed automated multisine input. This predictive test duly challenges the model by using data that were not included in the modeling process and that correspond to a different maneuver type. Quantitative validation results for longitudinal and lateral-directional motion are shown in Tables 7 and 8 for the square wave 3-2-1-1, and Tables 9 and 10 for the independent multisine input.

Table 7 Longitudinal model validation metrics comparison (3-2-1-1 maneuver)

Model basis	V_t	α	q	θ	a_x	a_z	Average
<i>Goodness of fit (GOF): 1 indicates perfect model fit</i>							
Pilot doublets	0.999	0.977	0.917	0.982	0.788	0.850	0.919
PTI doublets	0.999	0.980	0.946	0.990	0.799	0.919	0.939
Remote UPI	0.999	0.920	0.936	0.994	0.195	0.852	0.816
PTI multisines	1.000	0.956	0.962	0.997	0.760	0.961	0.939
<i>Theil inequality coefficient (TIC): 0 indicates perfect model fit</i>							
Pilot doublets	0.004	0.087	0.152	0.055	0.191	0.167	0.109
PTI doublets	0.004	0.080	0.122	0.040	0.186	0.118	0.092
Remote UPI	0.005	0.161	0.125	0.031	0.397	0.143	0.144
PTI multisines	0.002	0.125	0.104	0.023	0.195	0.078	0.088

Table 8 Lateral-directional model validation metrics comparison (3-2-1-1 maneuver)

Model basis	β	p	r	ϕ	a_y	Average
<i>Goodness of fit (GOF): 1 indicates perfect model fit</i>						
Pilot doublets	0.884	0.899	0.889	0.939	0.883	0.899
PTI doublets	0.952	0.935	0.953	0.965	0.937	0.948
Remote UPI	0.910	0.944	0.920	0.963	0.917	0.931
PTI multisines	0.949	0.946	0.956	0.973	0.924	0.950
<i>Theil inequality coefficient (TIC): 0 indicates perfect model fit</i>						
Pilot doublets	0.206	0.185	0.174	0.083	0.196	0.169
PTI doublets	0.121	0.144	0.107	0.062	0.135	0.114
Remote UPI	0.164	0.131	0.136	0.064	0.145	0.128
PTI multisines	0.114	0.128	0.098	0.054	0.138	0.106

Table 9 Longitudinal model validation metrics comparison (independent multisine)

Model basis	V_t	α	q	θ	a_x	a_z	Average
<i>Goodness of fit (GOF): 1 indicates perfect model fit</i>							
Pilot doublets	0.905	0.828	0.900	0.916	0.715	0.876	0.857
PTI doublets	0.893	0.778	0.931	0.945	0.746	0.930	0.870
Remote UPI	0.904	0.735	0.899	0.957	0.014	0.862	0.728
PTI multisines	0.943	0.684	0.971	0.993	0.736	0.952	0.880
<i>Theil inequality coefficient (TIC): 0 indicates perfect model fit</i>							
Pilot doublets	0.024	0.214	0.161	0.107	0.154	0.134	0.132
PTI doublets	0.025	0.239	0.131	0.088	0.146	0.100	0.121
Remote UPI	0.024	0.263	0.149	0.077	0.268	0.130	0.152
PTI multisines	0.018	0.297	0.085	0.032	0.148	0.081	0.110

Table 10 Lateral-directional model validation metrics comparison (independent multisine)

Model basis	β	p	r	ϕ	a_y	Average
<i>Goodness of fit (GOF): 1 indicates perfect model fit</i>						
Pilot doublets	0.754	0.901	0.727	0.967	0.731	0.816
PTI doublets	0.850	0.940	0.832	0.979	0.831	0.887
Remote UPI	0.759	0.912	0.702	0.980	0.765	0.824
PTI multisines	0.890	0.961	0.896	0.979	0.915	0.928
<i>Theil inequality coefficient (TIC): 0 indicates perfect model fit</i>						
Pilot doublets	0.283	0.173	0.319	0.094	0.324	0.238
PTI doublets	0.208	0.128	0.232	0.073	0.238	0.176
Remote UPI	0.258	0.152	0.292	0.070	0.265	0.207
PTI multisines	0.166	0.100	0.168	0.071	0.154	0.132

The average GOF values for all four input types is reasonably close to 1, indicating a good fit for the data. Additionally, average TIC values below 0.25 for all input types further demonstrate adequate models. Of particular note for this effort is that the model developed from remote UPI inputs showed model validation metrics comparable with the model validation metrics for models developed from automated doublets and multisine inputs. In most instances analyzed, models developed from automated inputs had superior model validation metrics to models developed from the respective counterpart piloted inputs; however, this disparity may be improved with increased test pilot proficiency. The similarity in model predictive results validates the utility of the UPI technique as a tool for unmanned aircraft aerodynamic modeling.

For qualitative analysis, the predictive capability for the model derived from each input type is assessed by comparing select model output variables to the collected flight data and with the Federal Aviation Administration (FAA) tolerance guidelines presented in

Ref. [33]. In general, these tolerance guidelines are $\pm 10\%$ of body axis rotation rates, $\pm 1.5^\circ$ of pitch attitude, and ± 0.1 g of normal acceleration.

Figures 13 and 14 show predicted output responses for the various developed models for validation maneuvers, along with the smoothed flight data and FAA tolerance limits. There is relatively good matching between the flight data shown in the solid green line, with the developed models shown in the dashed lines. The models do not consistently fall within the FAA tolerance criteria; however, this metric is intended for use with manned, large-scale aircraft. It is presented here for qualitative analysis and as an objective target for model fidelity.

D. Discussion of Input Types and Model Results

In general, model predictive performance is best for the model developed using multisine excitations and is worst for the model developed using the pilot doublet excitations. Quantitatively, the

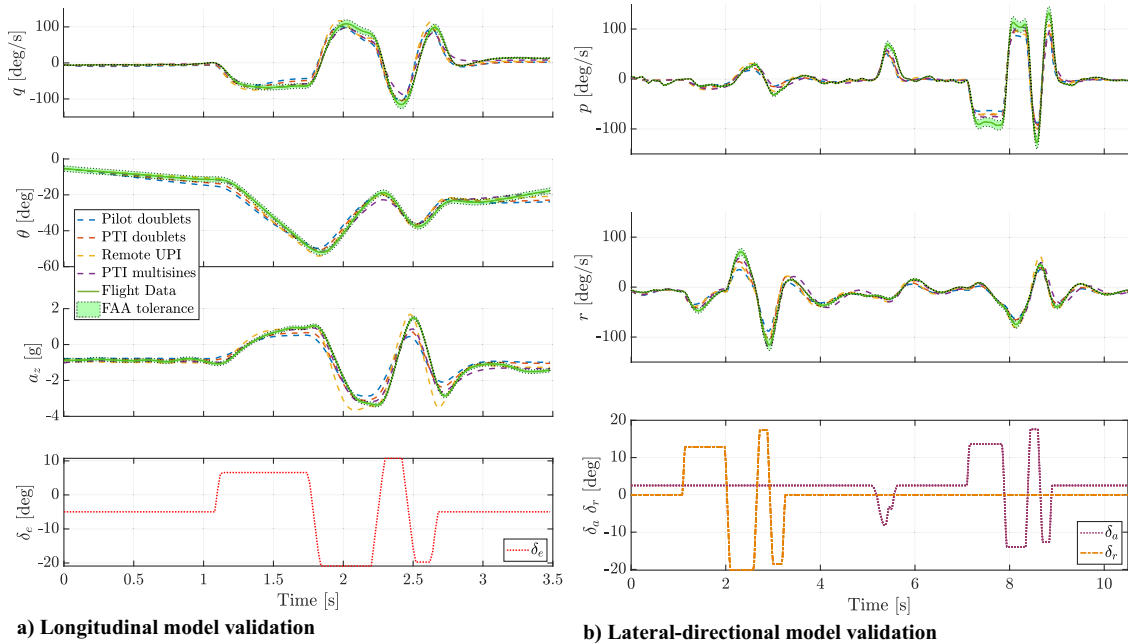


Fig. 13 Validation results for models developed from various inputs for 3-2-1-1 maneuver.

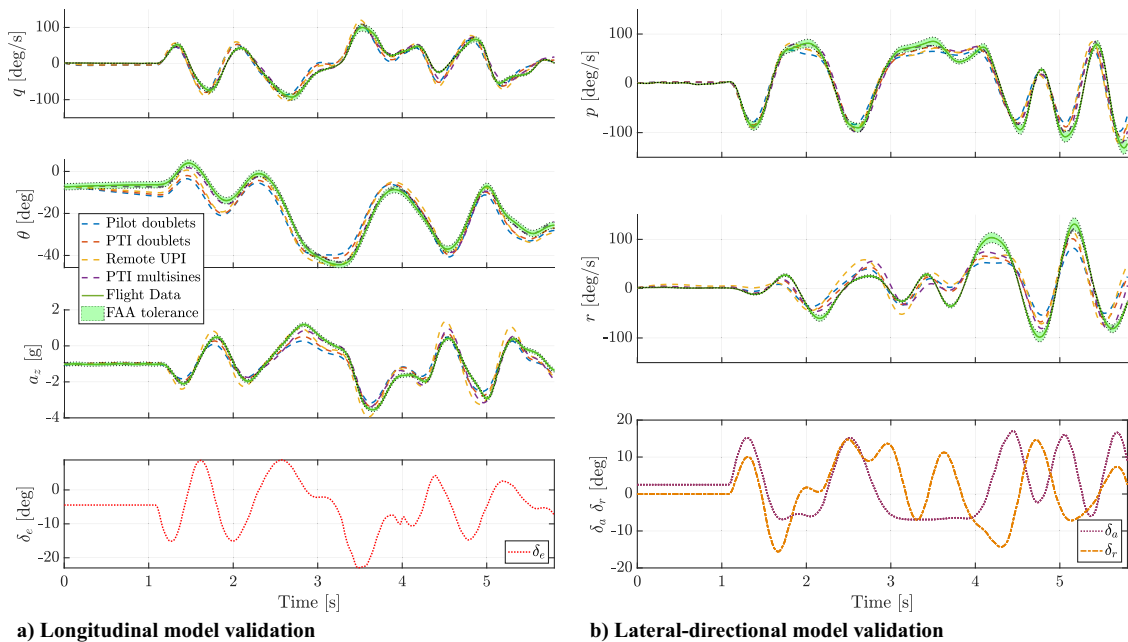


Fig. 14 Validation results for models developed from various inputs for independent multisine.

model generated from multisine inputs has the highest overall average for GOF (0.924) and the lowest overall average for TIC (0.109) for the results shown in Tables 7–10. A subjective qualitative assessment of model predictive results shown in Figs. 13 and 14 shows that the multisine generated model most closely predicts the flight data for independent maneuvers. All models show good matching in the modeling and validation stages, and the remote UPI-generated model prediction accuracy was comparable with models generated using the classic doublet and the computer-generated multisine excitation techniques.

The pilot doublet technique offers the simplest excitation technique with a minimal amount of training required to properly perform the input. Considering that this technique provides acceptable results, it is well suited for simple modeling efforts and preliminary analysis. The PTI doublet technique requires some engineering effort to generate automated inputs, but the coding to generate doublet inputs is relatively simple to implement. The pilot training required to execute a proper automated doublet is the lowest of any of the four input types, as the pilot simply trims the aircraft and activates a momentary PTI. This excitation technique provides repeatable, high-quality modeling results with relatively small confidence intervals, especially for an effort aiming to characterize the small perturbation dynamics in the heart of the aircraft envelope. The PTI multisine technique also requires automated inputs, and the engineering effort for implementation is more complex than for automated doublets. For a UAV with an unstable phugoid mode, as was used for this paper, the pilot must remain engaged during the maneuver to ensure that the aircraft does not stray from the desired modeling region. The multisine excitation technique provides high-quality modeling results very efficiently, and it is an excellent excitation technique beyond the linear aerodynamic region. The remote UPI technique requires the most pilot training and experience of any input type in this comparison. This technique provides high-quality modeling results and, importantly, may be used in nonlinear aerodynamic regions. The remote UPI technique also provides excitation very efficiently without the complications of incorporating computer-generated inputs. Each input type has benefits and drawbacks that lend their use to specific modeling objectives and programmatic considerations.

V. Conclusions

This investigation compared system identification input maneuvers and demonstrated the viability of remote UPI for unmanned aircraft by comparing validation results from a model generated using remote UPI and models developed with other input types. A simulation training tool was developed to improve remote UPI inputs, and flight test data showed that the training was useful. A nonlinear flight dynamic model for the My Twin Dream fixed-wing, unmanned aircraft was developed. Results indicate good matching between measured flight data and estimated model response for all maneuver input types. Furthermore, the model developed from remote UPI inputs showed agreement with validation maneuvers comparable to models developed from piloted doublets, automated doublets, and automated multisine inputs. This validated the remote UPI technique for unmanned aircraft aerodynamic modeling from flight test data.

Additionally, automated inputs were demonstrated using a cocomputer for small-aircraft aerodynamic model development. The validation results from models developed using automated inputs were comparable with validation results from models developed using piloted inputs. Remote UPI maneuvers are a good, low-cost alternative to automated inputs for unmanned aircraft flight test efforts that prioritize cost or programmatic speed, and include a proficient test pilot. On the other hand, automated multisine inputs produce superior modeling results with minimal pilot training. Automated multisines are also highly efficient, straightforward to apply in unusual flight conditions, and relatively safe because of low pilot workload, minimal impact to the flight trajectory, and precise control over the excitation input frequency content and amplitudes.

Acknowledgments

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